

Statistical Learning and Data Mining Assignment II

Analysis of App store data with Recommender Algorithm

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Data Description

The data contain information of about 17000 apps with their **ID**, **Name** and **Developer** of the App store. The dimensions of the data are **17007x13**. There are some numerical variables - **Average User Rating**, **User Rating Count**, **Price** and **Size of the app** and some categorical variables - **Primary Genre**, **Age Rating**. There are other variables like **Languages** supported by the app, **Secondary Genres**, **Release Date** and **Current Version Date**. Variables in Date format are converted in Year as knowing the exact date is not necessary. There are some duplicate entries in the data, which are deleted.

At the first glance, we can see many missing values in the data. After further inspection, we have found that almost all the missing observations are from two columns (variables) – **Average User Rating** and **User Rating Count**. This is clearly evident from the figure 1.

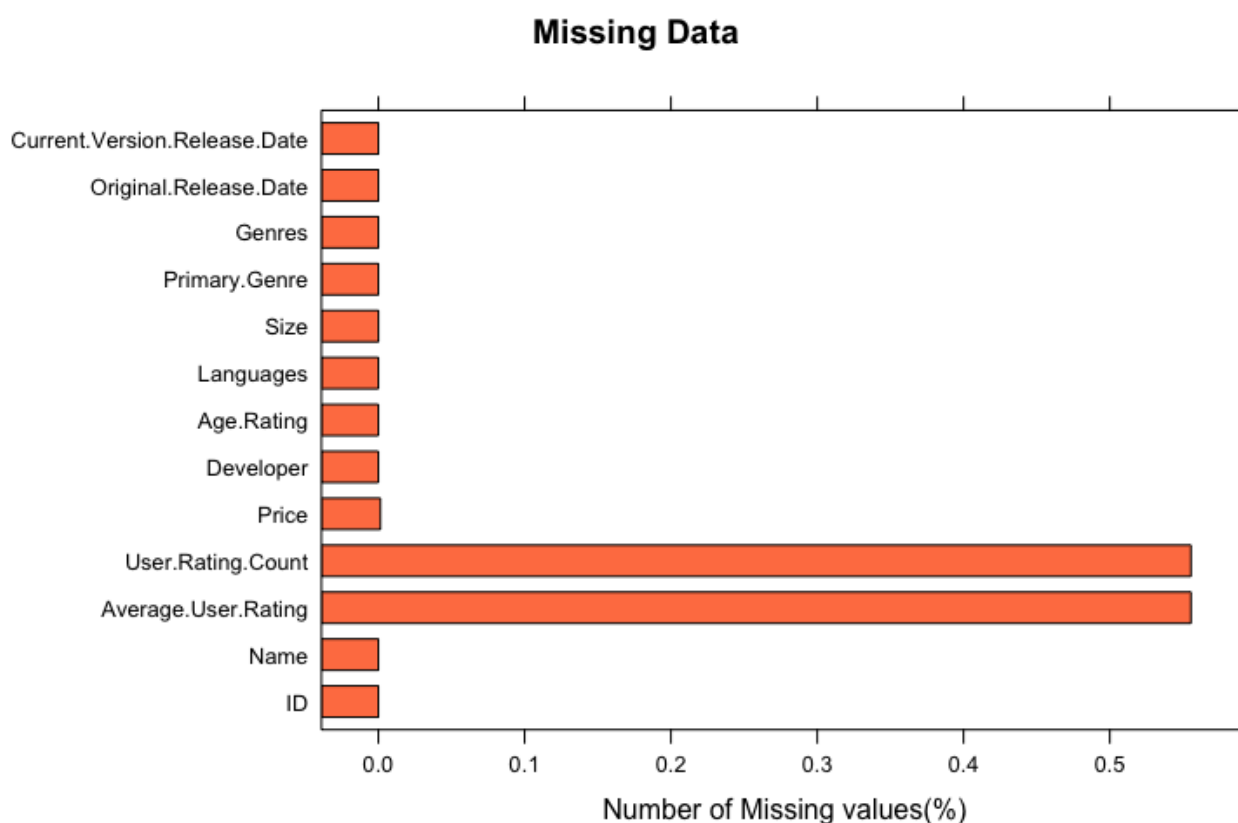


Figure 1

Also, proportion of missing values is very high (about 55%). Hence, we need to be very careful while selecting the imputation techniques. We will discuss about missing data and imputation in detail in the next section.

Objectives

There are four primary objectives.

- 1) To develop a popularity measure for the apps.
- 2) To build a Recommender algorithm for suggesting similar free apps for a required paid app.
- 3) To rank the genres according to popularity in the 17+ Age group.
- 4) To find some interesting findings from this data set using some statistical techniques.

Missing Data Imputation

There are two ways to handle the missing data.

- i) Deleting the Missing observations
- ii) Imputing the Missing observations

Missing observations can be deleted if they are small in number (5-10%) and only if we are sure that those values are MCAR (Missing Completely at Random). Here, about 55% are missing. Hence deleting those many observations will cause a huge loss of information. We need to choose a suitable technique to predict the missing values using the available data.

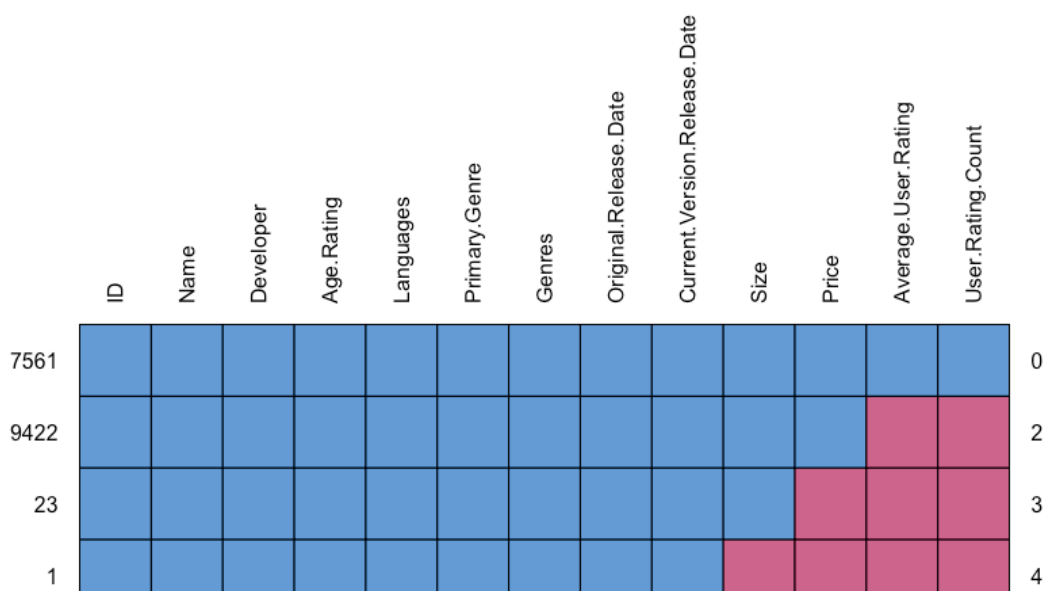


Figure 2

This plot indicates the missing values pattern in the Data. Both Average User Rating and User Rating Count contain **9422 missing observations**, and all of these occur together.

Note that, there are 23 missing observations in Price column and 1 missing observation in Size column. As these observations are very less in number, we omit these 24 observations. Here onwards, we will consider only 16993 observations for the Analysis.

We will now impute Average User Rating first and then using that column as one of the predictors, we will do the imputations for User Rating Count.

Random Forest Regression

Random forest is a Supervised Learning algorithm which uses ensemble learning method for classification and regression. The trees in random forests are run in parallel. There is no interaction between these trees while building the trees. It operates by constructing a multitude of decision trees at training time and outputting the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees.

We will use Random Forest Regression by treating Average User Rating as a response. The algorithm is tuned by 10-fold CV error by using ‘Caret’ package in R. Note that, the values of Average User Rating in all non-missing observations are 1.0, 1.5,... 4.5, 5. Hence, we need to round the predicted responses to nearest multiple of 0.5.

We have considered **6 variables** as regressors – **Price, Age Rating, Size, Primary Genre, Release Date and Current Version Release Date.**

The data with all the non-missing observations is considered as the training data. 10-fold CV is applied to this data and parameter ‘mtry’ is tuned. Predictions on the missing observations are made. **10-fold CV Error is 0.551766.**

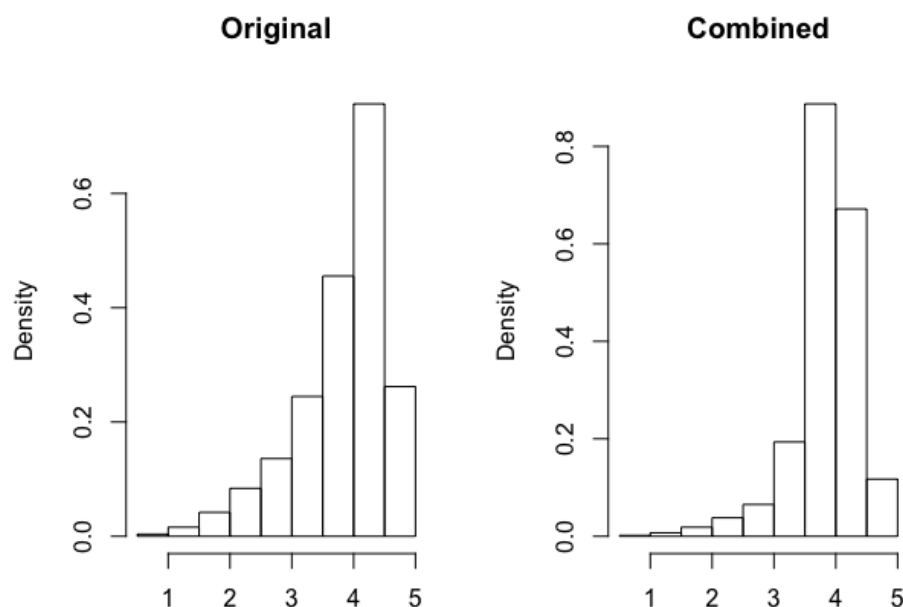


Figure 3

	MIN	Q1	MEDIAN	Q3	MAX	MEAN
NON-MISSING	1.0	3.5	4.5	4.5	5.0	4.061
WITH IMPUTATION	1.0	4.0	4.0	4.5	5.0	4.087

Table 1

There is not much difference in the distribution of Non-Missing observations and the Combined Data (Non-Missing + Imputed). From Table 1, we can see that summary statistics do not differ much for the two datasets. Hence, we conclude that our **imputation is fairly acceptable**.

Now, we will proceed for imputation of the missing observations from User Rating Count variable. For this, we have considered **7 variables** as regressors - **Price, Age Rating, Size, Primary Genre, Release Date, Current Version Release Date and Average User Rating**.

For Average User Rating, previously imputed observations are used along with all non-missing observations. Note that, one observation is removed because of very high value of count as this can cause unnecessary inflation in the summary statistics.

Summary Statistics of the Non-Missing and the combined data is provided below.

	MIN	Q1	MEDIAN	Q3	MEAN	MAX
NON-MISSING	5	12	46	308.2	2905.8	1277095
WITH IMPUTATION	5	33	88	283	1564	1277095

Table 2

As the values are very high, this much difference in the summary statistics is acceptable to us. Hence, **we conclude that this imputation is decent**.

For all the further analysis we will use the data with imputation which have **16822 observations**.

Exploratory and Descriptive Analysis

1) Free vs Paid Apps

To have successful app is the main goal of every developer. There is always an ambiguity in amateurs whether they should charge for their app or not. There are some advantages to a Paid app, extra revenue being the obvious one. Free apps make it easier to get tons of users.

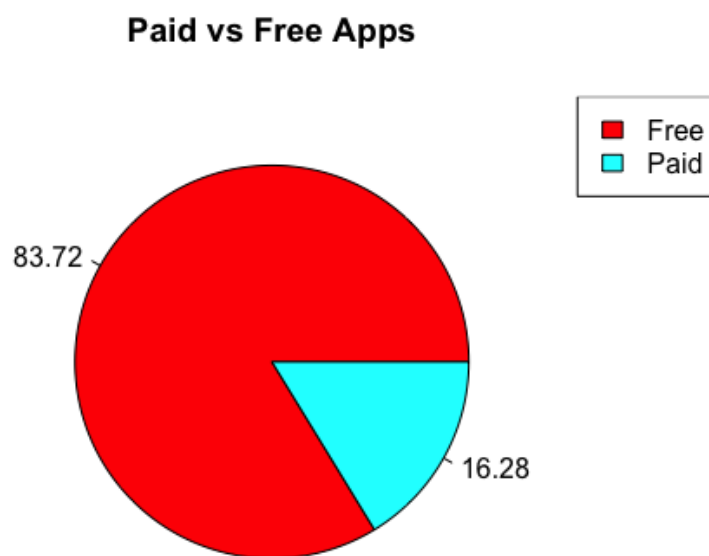


Figure 4

As expected, **Free apps constitute about 84% of the apps** in App store.

Which genres have more free apps?

Primary Genre	No. of Free Apps	% of Free Apps
Games	13571	84.24
Entertainment	170	86.29
Education	131	60.37
Utilities	57	74.03
Sports	50	83.33
Stickers	5	17.24
Other	99	75.00

Table 3

Stickers have very large proportion of paid apps. While the idea of basing your business's income on the sale of glorified emoticons seems a bit silly at first, the

fact of the matter is that stickers are helping some messaging apps bring in tens of millions of dollars each month.^[1]

Do the Paid apps have more user ratings?

Two sample t-test for equality of means is applied for the two groups – Free Apps and Paid Apps to see whether Means of Avg. User Rating differ significantly or not.

Our H_0 : Two means do not differ significantly gets rejected strongly (p-value is close to 0). We also tested H_0 against one sided hypothesis to see which group mean is larger. We have found that **Mean Avg. User Rating for Free apps is significantly greater than that of Paid apps**. This is surprising at first glance, but by going through some online literature we came to know that with free Apps, users tend to have lower expectations. Let's say they download your App, use it for a little while and but it isn't helping them in their everyday life. In the long run, what have they lost? Nothing! Hence, chances of giving negative ratings is very less for Free app users than Paid app users.

Welch Two Sample t-test

```
data: Free$Average.User.Rating and Paid$Average.User.Rating
t = 4.7735, df = 3789.6, p-value = 9.395e-07
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 0.03881187      Inf
sample estimates:
mean of x mean of y
 4.099020  4.039796
```

Summary 1

^[1] Kyle Russell, Business Insider India, October 2013

2) Genre wise distribution of Apps

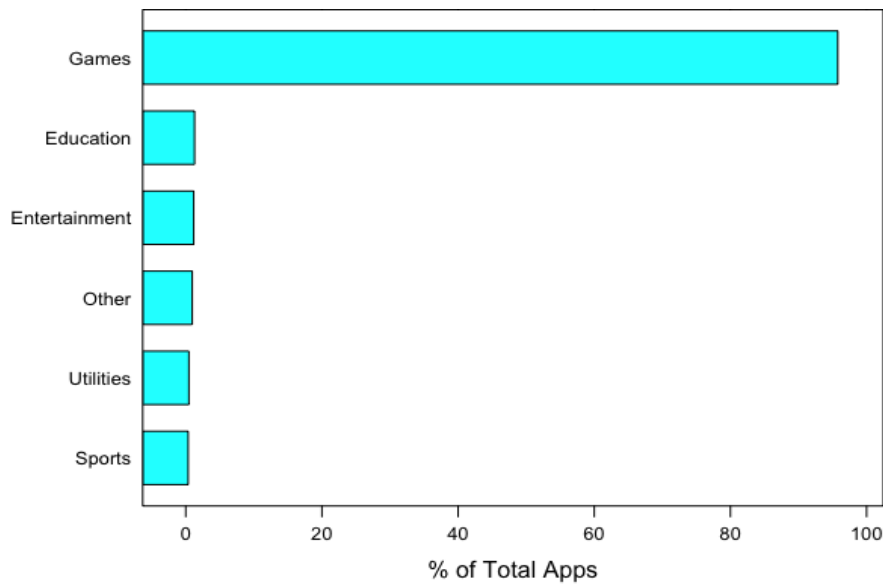


Figure 5

There are 21 unique primary genres in the data. As expected, number of Games are huge compared to other genres (See Figure 5). In terms of Avg. User Rating, Shopping and Music genres have highest mean but the number of apps in these genres are very low. **Games, which consists of more than 95% of App store apps has mean avg. user rating 4.1.**

Mean Avg User Rating by Genre

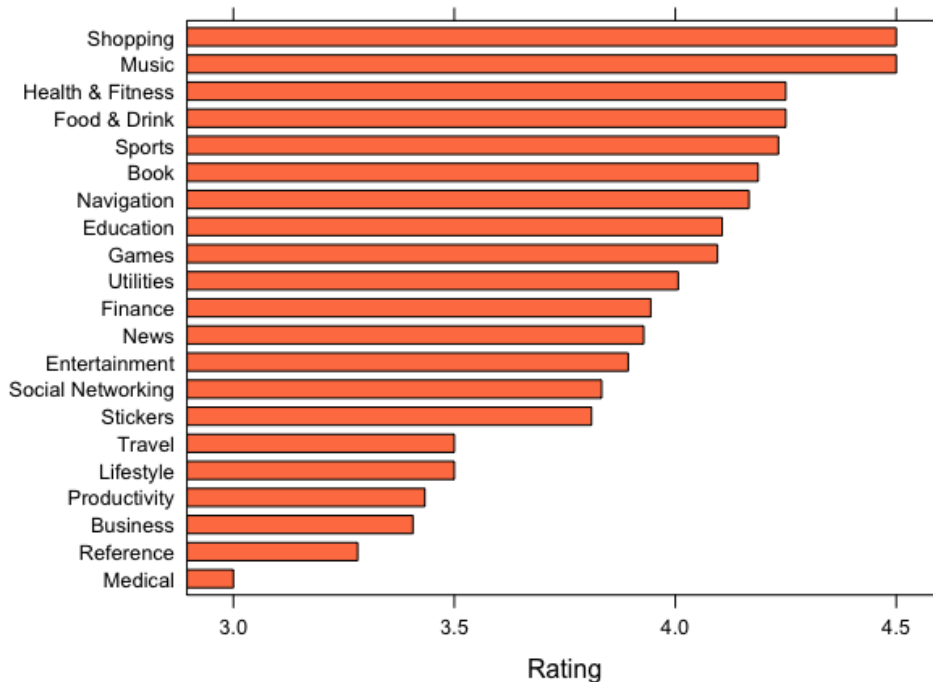


Figure 6

3) Age Rating wise distribution of Apps

Age Rating	4+	9+	12+	17+
% of Apps	69.47	14.58	12.06	3.89
Mean Rating	4.10	4.09	4.10	3.9

Table 4

There are four unique Age ratings in the dataset – **4+,9+,12+ and 17+**. As expected, **most of the apps have 4+ age rating**. Mean Avg. User rating for each rating group differ significantly (See ANOVA output). **Mean is lowest for 17+ Age group and it is significantly different from all the other groups**. The reason for this may be lies in the fact most of the people who use 17+ rated apps are matured and have large expectations from those apps.

```
              Df Sum Sq Mean Sq F value Pr(>F)
Data$Age.Rating  3      26    8.583   25.68 <2e-16 ***
Residuals      16818   5621    0.334
```

Summary 2

4) Effect of Size on the Avg. User Rating

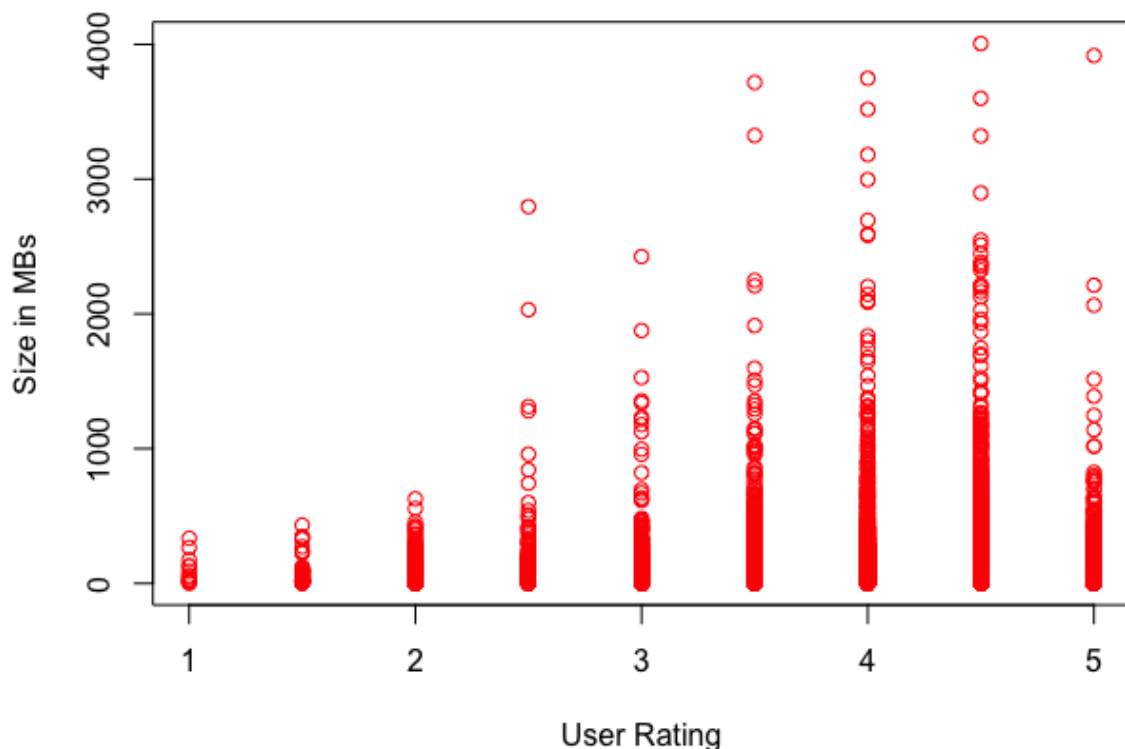


Figure 7

There is no direct relationship between Avg User Rating and App size. Correlation between them is **0.039, which is very low**.

Do the Paid apps have a larger size?

We have carried out two sample t-test for testing whether mean app size for Free and Paid apps differ significantly or not. **The alternative hypothesis that Paid Apps have larger size than Free apps is get strongly accepted** (See Summary 3). This is because the constant addition of new features and updates contributes a great deal to the size increase of many apps and for paid apps, updates and improvements are more frequent than free apps.

```
Welch Two Sample t-test

data: Free$Size/10^6 and Paid$Size/10^6
t = -6.5072, df = 3058.8, p-value = 4.457e-11
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
 -Inf -30.24582
sample estimates:
mean of x mean of y
 108.7836  149.2653
```

Summary 3

App Popularity Index

The quality of an app would be related to the user satisfaction, which could be measured by its popularity in App Store. There are various ranking algorithms which use lot of features to compute popularity of an app. Awdren De Lima et al. [2016] have provided very good description of the results of a mapping study that identified and analysed how metrics on apps' popularity have been addressed in the technical literature. 18 metrics were identified as related to apps' popularity (users rating and downloads the most cited). Most of these metrics need various time series variables like App ratings, No of downloads, No of uninstalls over weekly basis.

Frank et al. [2012] showed that the assessment of App's reputation is not reliable if it is only based on the average of user ratings because average rating itself is an unreliable measure. So they suggested to combine the quantity of participated users in ratings with the average of ratings in order to obtain a fair measure about popularity of an App.

One crude way to develop such index is multiplication between Avg. User Rating and No of Ratings. This will give the overall popularity of app.

Illustration:

Suppose there are three apps A,B,C with Avg. User Rating as 4.5,3.5,5 respectively. Now if decide popularity only on these values, then App C would turn out to be most popular app among three.

Now, let No of Ratings for A, B and C are 10,12 and 5 respectively. Then if we define popularity as the multiplication between Avg. Rating and No of Ratings, then app B would turn out to be most popular.

Drawback:

There is obvious drawback in this method. As No of ratings has much wider range than Avg. User Rating, eventually this popularity index can only be decided by No of Ratings only. For rectifying this drawback, we need to scale these two variables.

Popularity Index:

To overcome the previous problem, we have to modify our index a bit. Instead of using direct multiplication, we would first scale the two variables. Min-Max scaling is a very efficient way of scaling the data into the range 0-1. Suppose X is a nx1 vector then,

$$\text{Scaled}(X_i) = (X_i - \min(X)) / (\max(X) - \min(X)) \quad i = 1, 2, \dots, n$$

Popularity Index is given by

$$P_i = \text{Scaled}(\text{Avg. User Rating}) * \text{Scaled}(\text{User Rating Count}) * 100$$

The range of this index is 0 to 100. If the app has maximum Avg. User Rating along with highest number of Rating Count then, that app would have index value 100.

The top 10 apps by this popularity index are following –

Name	Avg. User Rating	Rating Count	Free/Paid	Popularity	Primary Genre
Clash Royale	4.5	1277095	Free	87.5000	Games
PUBG MOBILE	4.5	711409	Free	48.7419	Games
Plant vs Zombies \u2122 2	4.5	469562	Free	32.1718	Games
Cash, Inc. Fame & Fortune Game	5	374772	Free	29.3454	Games
Boom Beach	4.5	400787	Free	27.4596	Games
Pok\xe9mon Go	3.5	439776	Free	21.5221	Games

Idle Miner Tycoon: Cash Empire	4.5	283035	Free	19.3918	Games
Star Wars\u2122: Commander	4.5	259030	Free	17.7431	Games
Agar.io	4.5	257852	Free	17.6664	Games
Star Wars\u2122: Galaxy of Heroes	4.5	240990	Free	16.5111	Games

Table 5

Clash Royale is the most popular app by distance. All the Top 10 apps are Games and all of them are Free. This is not surprising as majority of the smartphone users are teenagers or young adults, and they tend to play lot of games for time pass. Also, general tendency of people is to use free apps than to paid for something.

From Table 6, we can see that most of the apps have popularity index below 0.05.

Min	Q1	Median	Q3	Max	Mean
0	0.00158	0.00497	0.01733	87.5000	0.10229

Table 6

This shows that there is huge gap between Top and Bottom apps. This is because there are lot of apps by amateur developers, who are not even close to apps developed by big gaming and software companies in terms of Features, Customer support, Bug free environment etc. This finding is consistent with the Nan Zhong et al. [2013]. Their results suggest that Google Play is more of a "Superstar" market strongly dominated by popular hit products than a "Long-tail" market where unpopular niche products aggregately contribute to a substantial portion of popularity.

Clustering of Apps: Is Popularity different in every cluster?

Clustering the apps to find subgroups seems difficult to implement in this case as number of apps is very large. Calculating the distance matrix itself would take up a huge amount of time and storage as dimension would almost be 17000x17000. A solution to this is using the technique CLARA (Clustering Large Applications, (Kaufman and Rousseeuw 1990)) is an extension to k-medoids (PAM) methods.

Instead of finding medoids for the entire data set, CLARA considers a small sample of the data with fixed size and applies the PAM algorithm to generate an optimal set of medoids for the sample. The quality of resulting medoids is measured by the average dissimilarity between every object in the entire data set and the medoid of its cluster, defined as the cost function.^[1]

CLARA repeats the sampling and clustering processes a pre-specified number of times in order to minimize the sampling bias. The final clustering results correspond to the set of medoids with the minimal cost. The variables considered are Average Rating, User Rating Count, Size, Price and indicators for Age Rating.

The optimal number of clusters is the one for which silhouette separation (which is a measure of how close each point in one cluster is to points in the neighbouring clusters) is maximum. This following plot shows K=2 is the optimal choice.

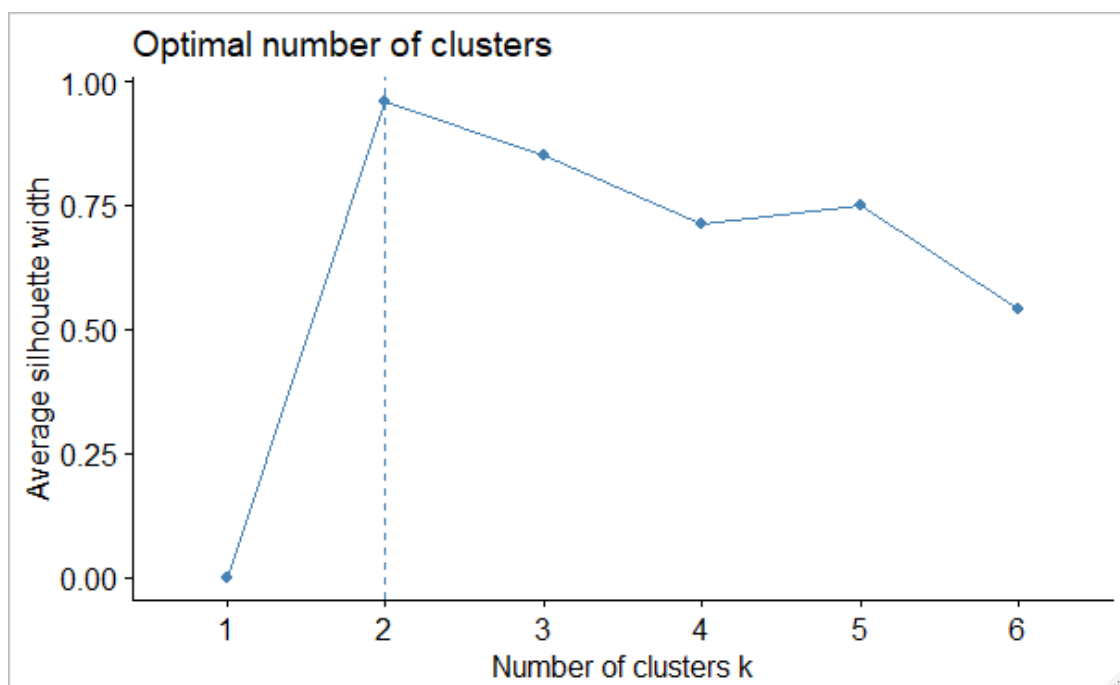


Figure 8

^[1] <https://www.datanovia.com/en/lessons/clara-in-r-clustering-large-applications/#clara-concept>

Two clusters are hence obtained and the medoids are as follows :

Average.User.Rating	User.Rating.Count	Price	Size
4.5	83	0	49.09056
4.5	22766	0	267.43910
Age.Rating_12+	Age.Rating_17+	Age.Rating_4+	Age.Rating_9+
0	0	1	0
0	0	1	0

Summary 4

The clusters differ drastically in User Rating Count and Size of the app. A visualization of the clusters in two dimensions as follows :

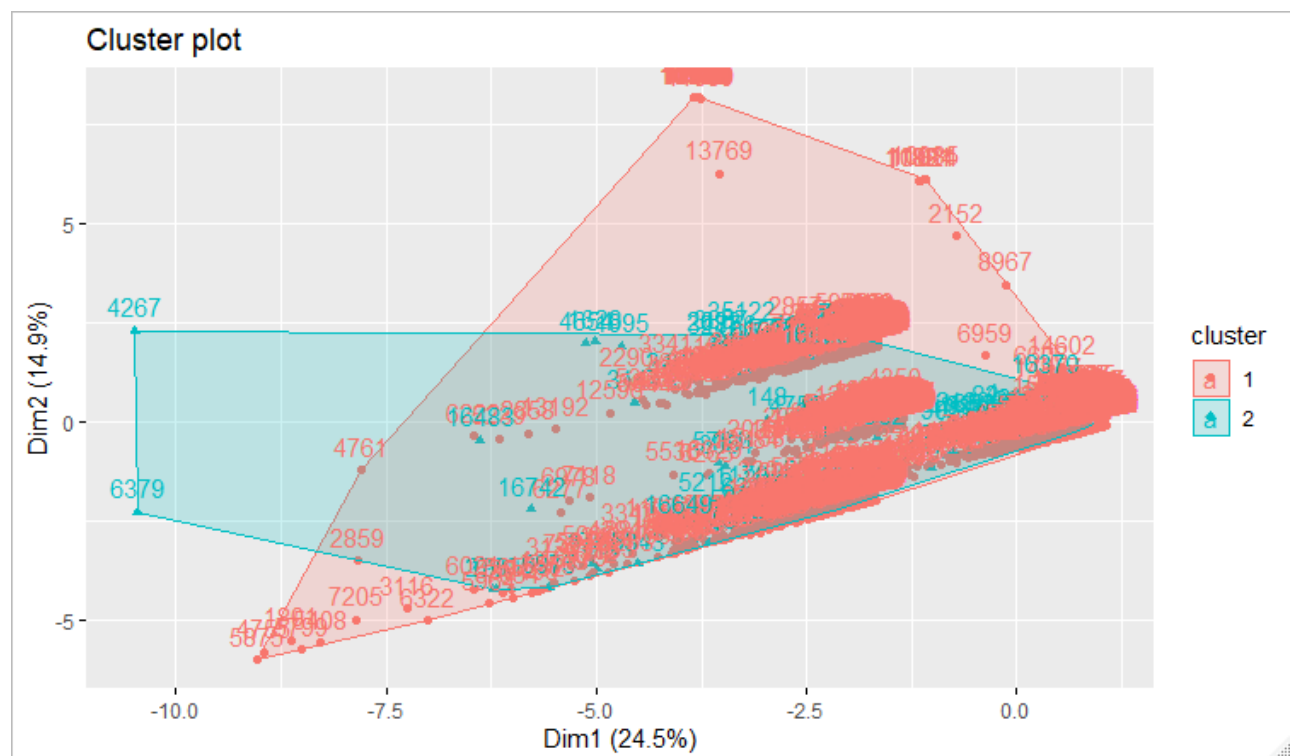


Figure 9

Taking into account the volume of the data, the above graph is not useful for observing the cluster memberships, but gives an idea about the clusters. The apps in Cluster 2 which seem to be outliers are :-

4263 - Clash Royale and 6379 - PUBG Mobile , which are the Top 2 in the most popular apps. (See Table 4)

The average popularity score in both the clusters is as follows :

	Cluster 1	Cluster 2
Average Popularity Score	0.02670953	3.491264

Table 7

We have carried out two sample t-test for testing whether average Popularity Score for both clusters differ significantly or not. The alternative hypothesis that the first cluster has lesser average Popularity than the second is strongly accepted. The summary for the test is as follows.

```
welch Two Sample t-test

data: Pop1 and Pop2
t = -10.172, df = 366, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
-4.134307 -2.794801
```

Summary 5

Thus we can say Cluster 2 includes the apps which are more popular.

All of the Top 100 Apps according to the Popularity Score belong to Cluster 2, while all the Bottom 100 belong to Cluster 1.

App	Popularity Score	Cluster
Plants vs. Zombies	32.171764	2
Cash, Inc. Fame & Fortune Game	29.345387	2
Boom Beach	27.459635	2
Spider Solitaire	0.4253420	1
Deep Stream	0.3731335	1
Galaxy Challenger	0.1856760	1

Table 8

This proves that though we clustered based on all the variables mentioned above and the Popularity Score is based only on Average Rating and User Count, it manages to capture the popularity of an app very efficiently.

Are Smaller apps more popular?

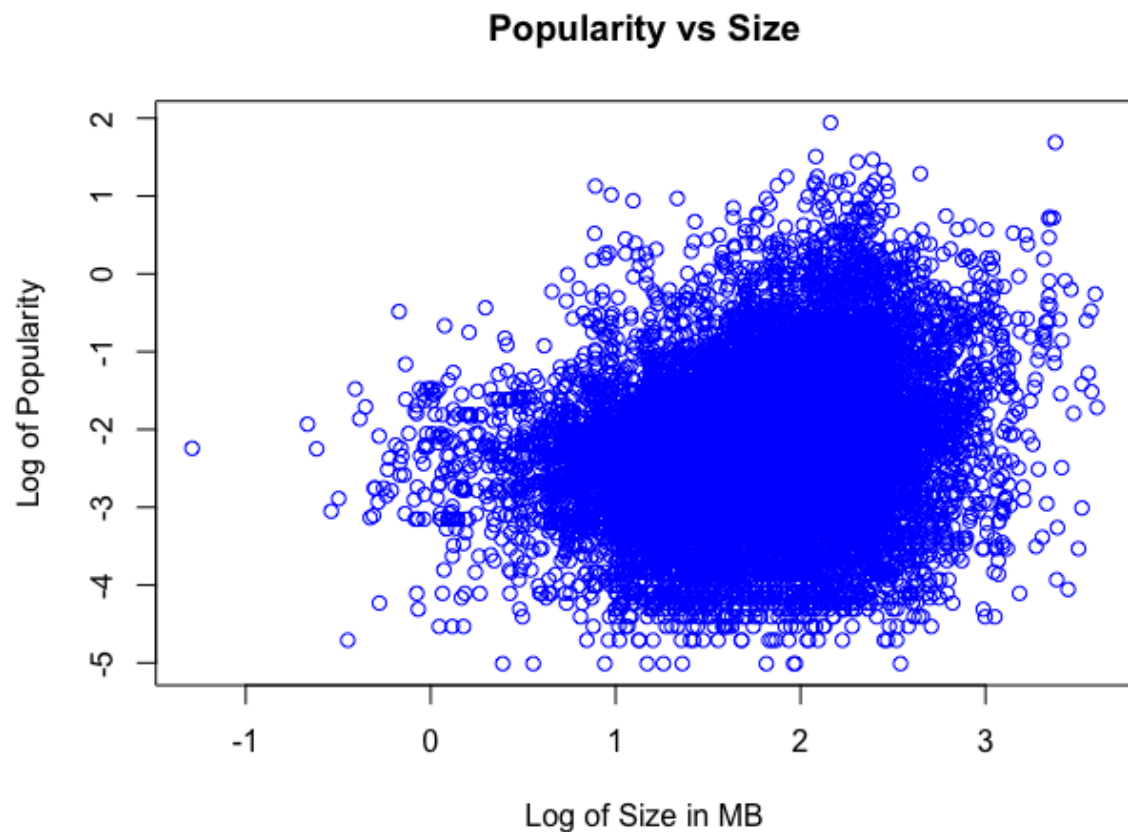


Figure 11

There is no direct relationship between App size and Popularity. Correlation Coeff. between them is 0.0799, which indicates very weak correlation.

Genre wise Popularity

First, we will look at the popularity of apps by primary genre.

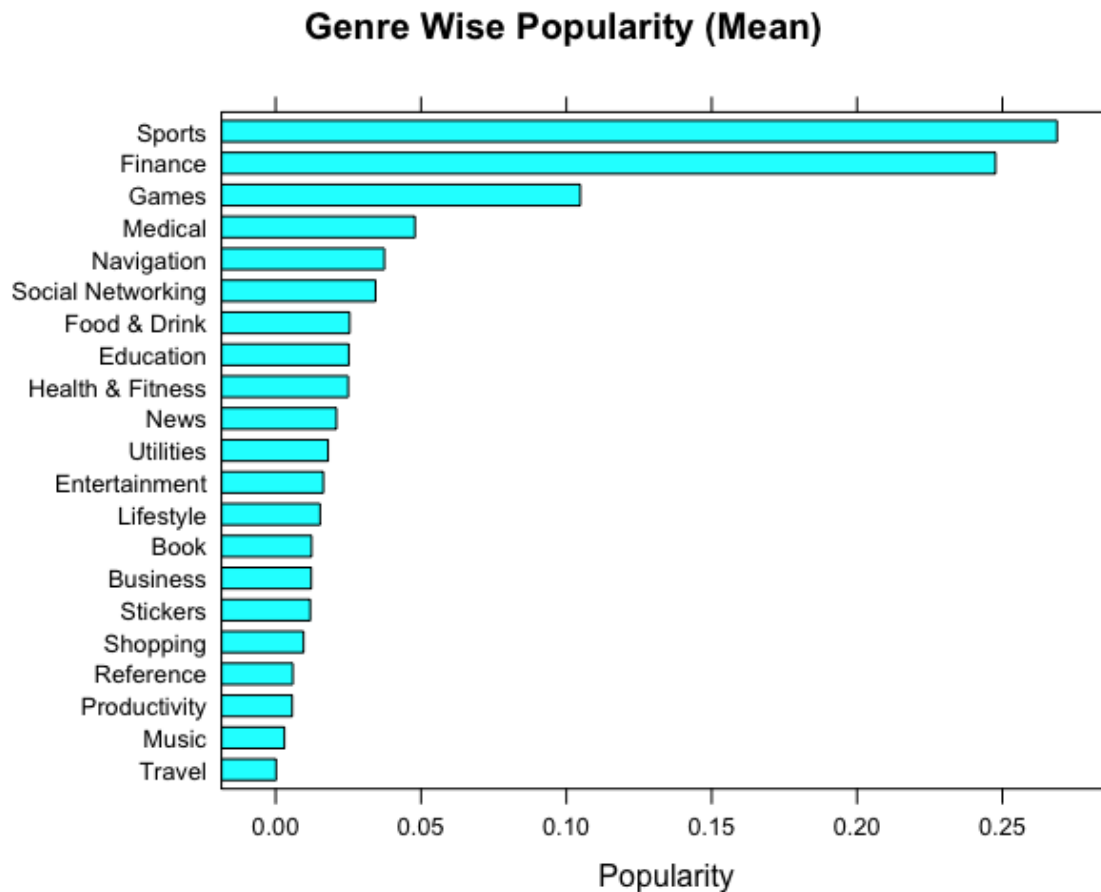


Figure 12

Sports is the most popular genre followed by Finance, Games and Medical. Travel is the least popular genre. But this doesn't seem fair as many of the genres have very low number of apps. For example, Travel has only one app, Medical has only two. Figure 12 gives Genre popularity of only those genres which have at least 50 apps.

Sports is the most popular and Entertainment is the least popular genre (See Figure 13). The main reason for worst performance of Entertainment genre can be attributed with miscategorization of apps, which we will discuss in detail in next section.

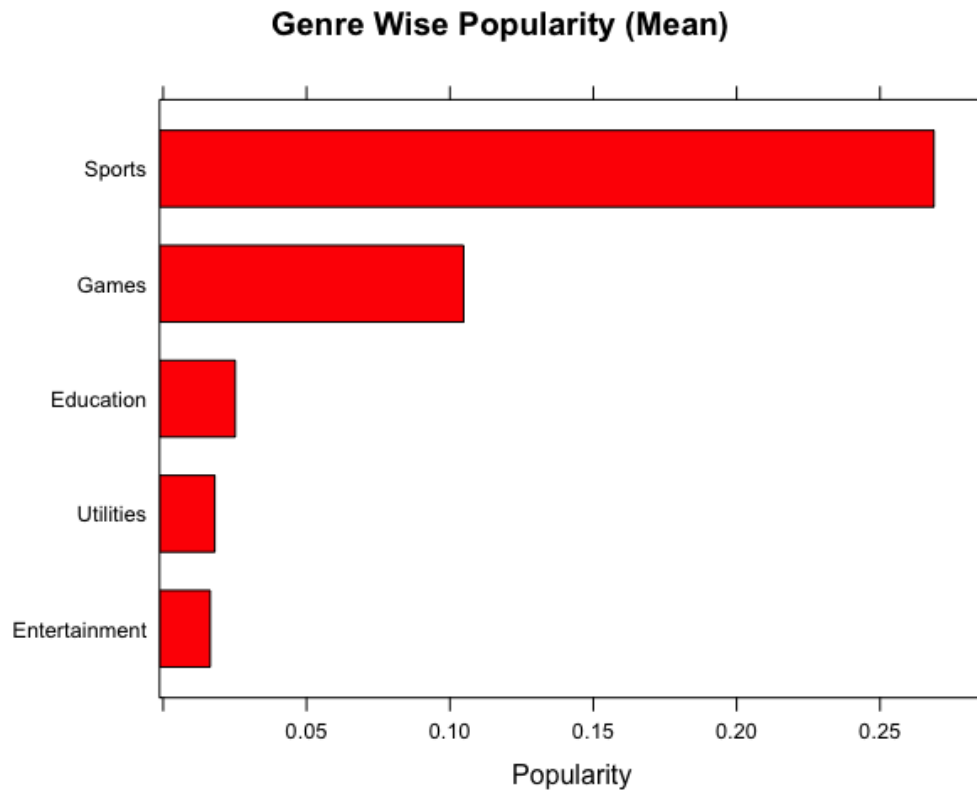


Figure 13

Now, we will consider another genre column i.e. secondary genre. Also, we have considered only **17+ age group** apps.

Music is the most popular genre followed by Cards while **Productivity is the least popular**. This seems reasonable as many Music videos enjoyed by young adults have 17+ Age rating and though card games have 17+ rating (Due to some regulations in many countries), they are played by all the age groups (See Figure 14).

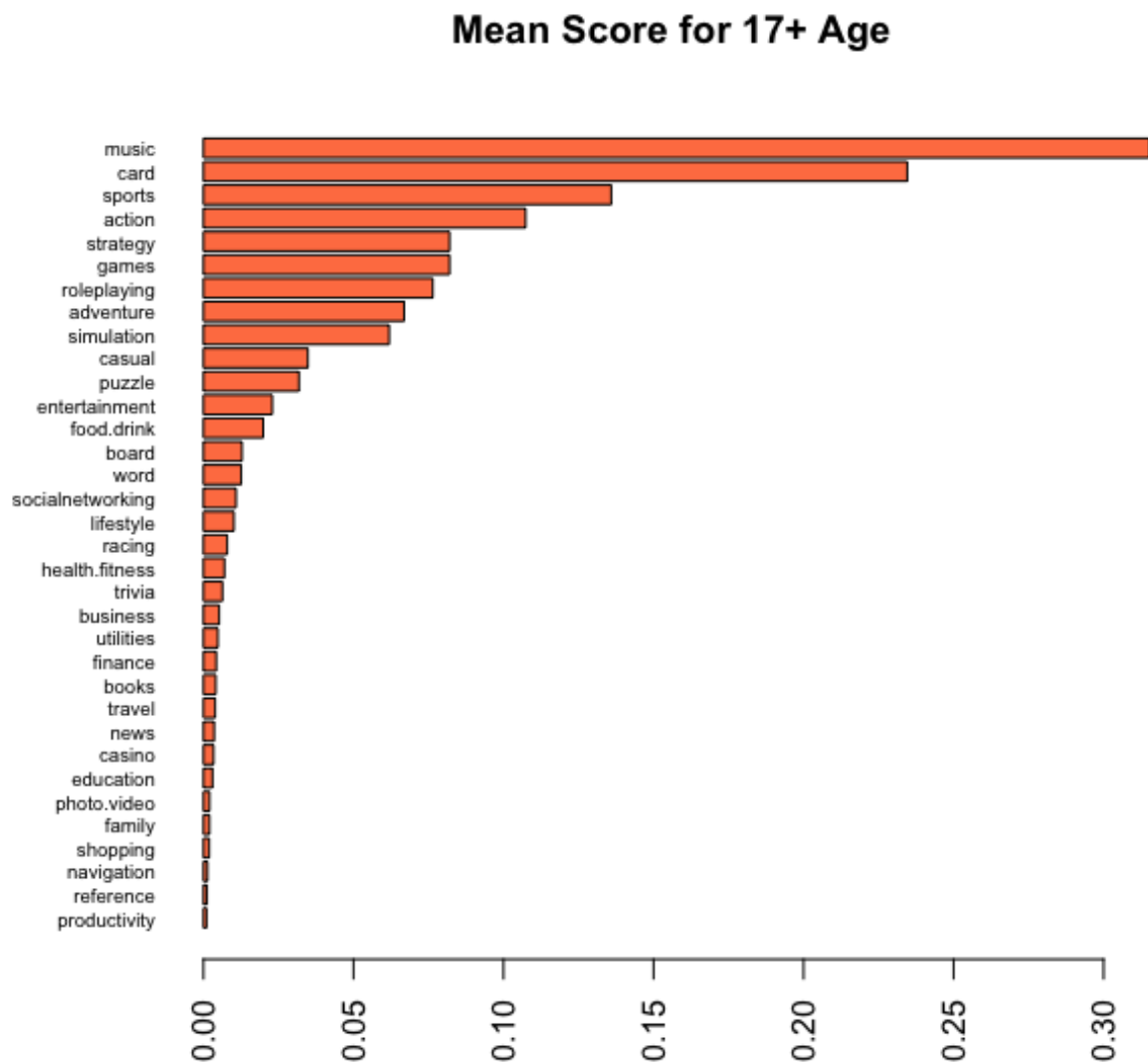


Figure 14

The problem of miscategorization is more prominent for this column. All the apps have games as a secondary genre, and all but one app have strategy as secondary genre.

For example, consider ID 719628242 : PDF Editor Pro - Take Note, Sign & Fill Forms, clearly this app has nothing to do with category games and strategy. Still it is classified in those.

This problem is explained in detail in the last section.

Recommender System

A recommender system is an information filtering system which generates meaningful recommendations to a collection of users for items or products that might interest them.

Our goal is to recommend the user an app that is free, which is similar to the Paid app they like.

Some variables that were created for analysis, described as follows.

1. Price indicator : 0 if the app is Free and 1 if Paid.
2. Genre count : Number of Secondary Genres
3. Indicator variables for Secondary Genres (Total 47 Genres) : 1 if present as Secondary and 1 if not.

Name	Age Rating	Developer	Primary Genre	Genre	Genre Count	Price indicator
Sudoku	4+	Mighty Mighty Good Games	Games	Games, Strategy , Puzzle	2	1
Sudoku (Free)	4+	Mighty Mighty Good Games	Games	Games, Strategy , Puzzle	2	0

Table 11

For every paid app, some temporary variables are created.

1. Developer indicator : 1 if developer of the free app is same as the Paid app, 0 if not.
2. Genre Match : Number of Secondary genres common to the Paid app and Free app divided by total number of Secondary genres in the Paid app. (1 if all present, hence more likely to be similar)

The first thing we notice about any app is its Name. The developer decides a name that explains what the app is about completely. A user hence knows the function of an app named “PDF scanner” or “Online Chess” without any ambiguity.

Thus, matching the keywords in the name contributes a major part in this system.

Keywords are the separated words in the name excluding the punctuation. Two new lists are created as follows.

1. Keywords for Free Apps (keywords_f)
2. Keywords for Paid Apps (keywords_p)

To avoid errors due to commonly occurring words like “of”, “the” or “game”, a list of general keywords is decided by observing the app names.

App Name	Keywords	Keywords excluding general keywords
Battle On The Sea for iPad (Paid)	Battle, On, The, Sea, For, iPad	Battle, Sea, iPad
Fleet Battle: Sea Battle game (Free suggested)	Fleet, Battle, Sea, Game	Fleet, Battle, Sea

Table 12

We observe in the above example, common keywords can lead us to a similar app.

The working of the Recommender system algorithm is as follows:

Partition the data as Paid Apps and Free Apps

For recommendation of a Free App for every Paid one, following algorithm is used

For every Paid App :

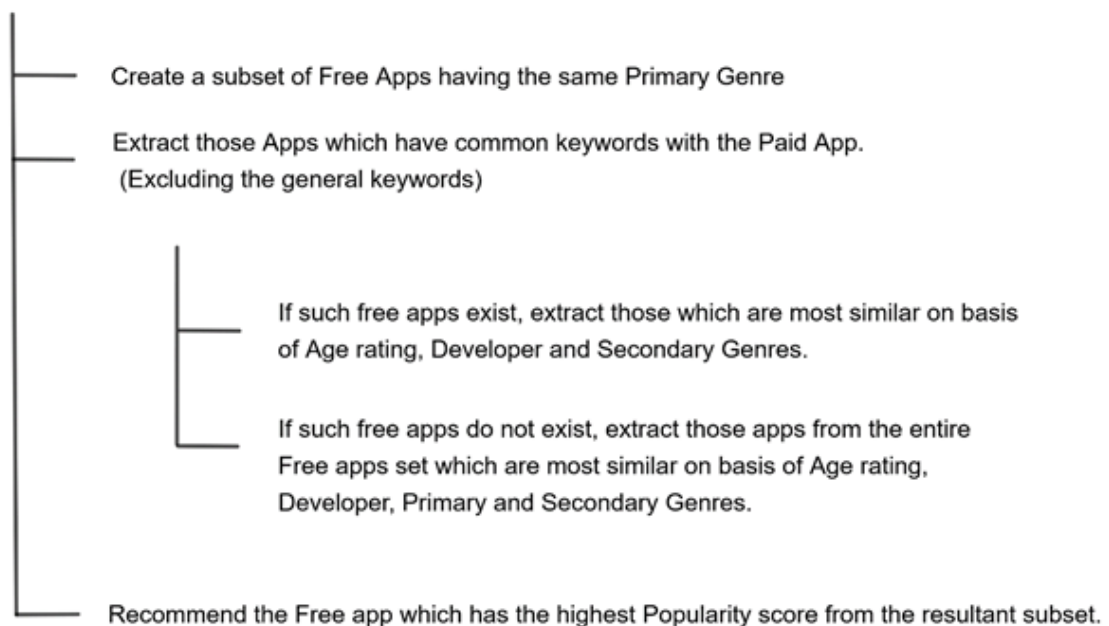


Figure 15

For computing the similarity between apps, Gower distance is used. Out of the total 2800 Paid apps, almost 2500 were worked out in the first part (where common keywords exist), with only about 300 in the second.

The recommended app must be age appropriate, and must be as close to the paid app genre. Developers tend to release Paid and Free versions of the same apps. Thus these points are taken into consideration for measuring similarity, and the variables used are Age rating, Developer indicator and Genre Match.

The variable Size is not considered for similarity as from Summary 3, Similar Paid and Free apps would not have similar sizes.

The most Popular app among the set of similar apps is selected as the final recommendation. Hence an app that has a higher user count and average user rating is more likely to be recommended.

Some apps with satisfactory recommendations

Paid App	Suggested Free App	Primary Genre
Yatzy for iPad	Super Yatzy - Six Dice of Fun!	Games
Zombie War : Apocalypse of Global Defense	Dead Ahead: Zombie Warfare	Games
Iron Heart: Steam Tower TD	Iron Sea Frontier Defenders TD	Games
Chess Clock Pro - Timer for your games	Chess Clock Timer	Utilities
Finabase: realtime stocks	Stock Master: realtime stocks	Finance
Vainglory Stickers	Iron Marines Stickers	Stickers
Commander English	English Words Game	Educational
Luzbel - Interactive Book app scary horror story	20000 Leagues Under the Sea - Interactive Fiction	Book
Football Team Builder Pro	Cricket Dream Team	Sports

Table 12

Some apps with flawed recommendations

Paid App	Suggested Free App	Primary Genre
IWO: Bloodbath in the Bonins	Backgammon+	Games
Documents Pro - for Microsoft Office Word edition	Magic Personal Life Counter	Utilities
LetsCode:Java	Education Puzzles	Education
Dominos and Bones by Konsole Kingz	Tic Tac Toe Stickers & Game	Stickers
Notepad Pro - Annotate PDFs, Notes Taker	Signalyx	Business
Axis & Allies 1942 - AA Tool	Diwali Rangoli Designs 2019	Productivity

Table 13

This point is explained in the last section. The misclassifications are observed to be due the wrongly matched keywords and incorrect classification into Secondary genres. Most of the misclassifications are in the non-games genre.

Though the whole system takes about 3.5 hours to run (on Windows 10 Intel(R)Core i3 (2GHz) 4 GB RAM), an individual recommendation takes about 4 seconds. Speed is an important aspect for a recommender system as a good system is able to offer a good recommendation as quickly as possible. This time will be further reduced if a machine with higher power is used.

App store Categorization: Is it any Good?

Genres are a major contributor to what makes an app discoverable. Listing secondary genres in addition to Primary genres is advised, as Primary genres by themselves tend to be very vast.

To observe patterns in the genres, we have used the terms from Association Rule Mining. Primary and Secondary genres for an app is treated as a transaction. The support and count for some genres is as follows:

items	support	count
{games}	1.000000000	16823
{strategy}	0.999940558	16822
{entertainment}	0.469892409	7905
{puzzle}	0.232895441	3918
{simulation}	0.126077394	2121
{action}	0.117993224	1985
{board}	0.100814361	1696
{casual}	0.100041610	1683
{roleplaying}	0.066397194	1117
{education}	0.055756999	938

Summary 6

Games and Strategy have support equal to 1. This is because every app is placed in categories Games, Strategy. Though 95% of the apps have Primary Genre as Games, most of the remaining apps are definitely not games. This indicates some developers tend to list their apps in “Games” as the Secondary Genre category to increase the reach.

On the contrary, some of the apps which are games are listed under “Entertainment” as Primary Genre, which has Average Popularity score as 0.0164, with “Games” as Secondary Genre which has Average Popularity score as 0.1048. This indicates may developers have avoided using “Games” as primary genre and have opted for a less competitive one, so as to gain an edge while recommendation.

Some examples are as follows :

Name	Primary Genre	Secondary Genres
Go Books	Book	Books, Games, Board, Strategy
Production Calculator Lite	Productivity	Productivity, Simulation, Games, Strategy
Office Docs - for MS Word editon	Business	Business, Games, Puzzle, Strategy

Stock Master: realtime stocks	Finance	Finance, Simulation, Strategy, Games
Scary Maze Game Lite	Entertainment	Entertainment, Strategy, Puzzle, Games
Solitaire On Vacation	Entertainment	Entertainment, Strategy, Card, Games
Whodunnit -Murder Mystery Game	Entertainment	Entertainment, Games, Strategy, Role Playing

Table 14

This miscategorisation is a source of annoyance to both, the platform and the users. Such miscategorisation is heavily punished by the Google and Apple play stores. All apps get reviewed manually by their teams, and apps that don't conform to their standard may get deleted.

This Miscategorisation problem for Google Play Store has been discussed in detail by Mihhail Matskin et al. [2013]. By employing a probabilistic topic modelling technique and K-means clustering method, they have found out that categorization system of Google Play does not respect properly similarity of applications.

Concluding Remarks

Increasing usage of smart phone apps and their easy availability has resulted in a plethora of choices in front of a user. Studies have found out that too many options results in user being overwhelmed, second guessing the decision or have trouble making a decision and end up buying nothing which is a phenomenon known as “Analysis paralysis.” Hence it is imperative to have a credible recommender system, which provides the user with a good alternative.^[1]

Research suggest that platforms like Google Play or Apple Store are more of a "Superstar" market strongly dominated by popular hit products than a "Long-tail" market where unpopular niche products aggregately contribute to a substantial portion of popularity. Blockbuster apps have more downloads, higher ratings and satisfaction ratio. Hence it is important to decide on a good measure of Popularity which takes into account the entire market.^[Ref 7]

Now we attempt to answer all the questions raised in the Objectives.

Popularity Index is developed using scaled average rating and user rating count. This index takes into account both, the reach of the app and the public consensus about the quality. The top 3 Most popular apps are:
Clash Royale, PUBG MOBILE and Plant vs Zombies

All these giants in the industry and are dominating the market. These apps have many advantages over amateur developers in terms of marketing budget, quantity and quality of engineers and brand hype. Hence at the present moment it is almost impossible for an amateur app to beat these blockbusters in terms of Popularity.

We built a Recommendation System which suggests a similar app for a desired Paid app. The recommendations are based on the Primary and Secondary Genre, Keywords in Name, Age rating, Developer and the App Popularity.

[1] <https://www.weebly.com/inspiration/paradox-of-choice-and-analysis-paralysis-limit-options-increase-sales/>

Top 10 Popular Paid Apps with their Recommended Free Apps

Paid App	Popularity Score	Recommended Free App
Plague Inc.	14.04622	Business Inc. 3D Simulator
Traffic Rush	10.41225	Traffic Rush 2
Bloons TD 5	7.655764	Bloons TD Battles
Boxed In	4.562619	Boxed!
Bloons TD 6	3.651191	Bloons TD Battles
Jungle Crash Land	3.311484	Smashers.io Foes in Worms Land
Bloons TD 5 HD	2.921965	Bloons TD Battles
TowerMadness	2.489096	TowerMadness Zero
Hotel Dash Deluxe	2.48107	Wedding Dash

Table 15

Popularity of a Genre in the age bracket 17+ is regarded as the average of all the apps which are listed under that genre, either as Primary or as Secondary Genre. The most popular Genres in this age category are: Music, Card and Sports. The least popular Genres in this age category are: Productivity, Reference and Navigation.

Interesting Findings from the data:

Miscategorization of Apps into genres to gain competitive edge is something that affects adversely the accuracy of analysis and recommendation systems across the platforms. Listing an app under a popular but unrelated genre to increase the reach or under a less popular genre to decrease competition is observed in some cases.

From our analysis, it is clear that App store suffers from this problem as many of the apps have placed in the wrong genres. (See Page 27-28)

Another interesting finding from the work is that Avg. User Ratings of Free apps is significantly greater than that of Paid apps. This is because users generally have lower expectations from free apps as opposed to the ones they invest money in. (See Page 9)

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